

ad Background	Viewport
Router 5: Se0/1/0: ip 192.168.5.2 Fa0/0: ip 40.40.40.1 PC11: ip 40.40.40.2 PC12: ip 40.40.40.3 PC13: ip 40.40.40.4 Serverftp2: ip 40.40.40.5 Router 6: Se0/1/0: ip 192.168.6.2 Se0/1/1: ip 192.168.7.1 Fa0/0: ip 50.50.50.1 PC16: ip 50.50.50.2 PC17: ip 50.50.50.3 DNSServer2: ip 50.50.50.4 WebServer2: ip 50.50.50.5 Router 7: Se0/1/0: ip 192.168.7.2 Fa0/0: 60.60.60.1 PC14: ip 60.60.60.2 PC15: ip 60.60.60.3 DNSServer1: 60.60.60.4 WebServer1: 60.60.60.5 Router 8: Se0/1/0: ip 192.168.8.2 Se0/1/1: ip 192.168.9.1 Fa0/0: ip 70.70.70.1 PC19: ip 70.70.70.2 DNSServer3: ip 70.70.70.3 EmailServer1: ip 70.70.70.4 PC20: ip 70.70.70.5	Router 9: Se0/1/0: ip 192.168.9.2 Fa0/0: ip 80.80.80.1 PC21: ip 80.80.80.2 DNSServer4: ip 80.80.80.3 EmailServer2: ip 80.80.80.4 PC18: ip 80.80.80.5

IPs Through DHCP to PCs

Router2

Physical Config CLI

IOS Command Line Interface

```
Cisco IOS Software, 1841 Software (C1841-ADVIPSERVICESK9-M), Version 12.4(15)T1,
RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 04:52 by pt_team

Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip DHCP excluded-address 10.10.10.2 10.10.10.5
Router(config)#ip DHCP POOL Hammadullah15398Assignment2
Router(dhcp-config)#network 10.10.10.0 255.0.0.0
Router(dhcp-config)#default-router 10.10.10.1
Router(dhcp-config)#exit
Router(config)#
```

Copy Paste

Router3

Physical Config CLI

IOS Command Line Interface

```
2 FastEthernet/IEEE 802.3 interface(s)
4 Low-speed serial(sync/async) network interface(s)
191K bytes of NVRAM.
63488K bytes of ATA CompactFlash (Read/Write)
Cisco IOS Software, 1841 Software (C1841-ADVIPSERVICESK9-M), Version 12.4(15)T1,
RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 04:52 by pt_team

Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip DHCP excluded-address 20.20.20.2 20.20.20.10
Router(config)#ip DHCP POOL Hammad15398Assignment2
Router(dhcp-config)#network 20.20.20.0 255.0.0.0
Router(dhcp-config)#default-router 20.20.20.1
Router(dhcp-config)#exit
Router(config)#
```

Copy Paste

Router 2 PCs
PC4, PC5, PC6, PC7

PC4

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address: 10.0.0.1

Subnet Mask: 255.0.0.0

Default Gateway: 10.10.10.1

DNS Server:

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address: /

Link Local Address: FE80::201:C7FF:FE5C:CEA8

IPv6 Gateway:

IPv6 DNS Server:

PC5

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address: 10.0.0.2

Subnet Mask: 255.0.0.0

Default Gateway: 10.10.10.1

DNS Server:

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address: /

Link Local Address: FE80::202:16FF:FE2E:4E0C

IPv6 Gateway:

IPv6 DNS Server:

PC6

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address 10.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 10.10.10.1

DNS Server

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:D3FF:FE25:B5C0

IPv6 Gateway

IPv6 DNS Server

PC7

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address 10.0.0.4

Subnet Mask 255.0.0.0

Default Gateway 10.10.10.1

DNS Server

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address FE80::204:9AFF:FE4B:D74B

IPv6 Gateway

IPv6 DNS Server

Router 3 PCs
PC0, PC1, PC2, PC3

PC0

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address: 20.0.0.1

Subnet Mask: 255.0.0.0

Default Gateway: 20.20.20.1

DNS Server:

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address: /

Link Local Address: FE80::202:17FF:FEAA:C516

IPv6 Gateway:

IPv6 DNS Server:

PC1

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address: 20.0.0.2

Subnet Mask: 255.0.0.0

Default Gateway: 20.20.20.1

DNS Server:

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address: /

Link Local Address: FE80::20D:BDFF:FEE5:B93D

IPv6 Gateway:

IPv6 DNS Server:

PC2

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address 20.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 20.20.20.1

DNS Server

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address FE80::201:C7FF:FE9E:6B9B

IPv6 Gateway

IPv6 DNS Server

PC3

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address 20.0.0.4

Subnet Mask 255.0.0.0

Default Gateway 20.20.20.1

DNS Server

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

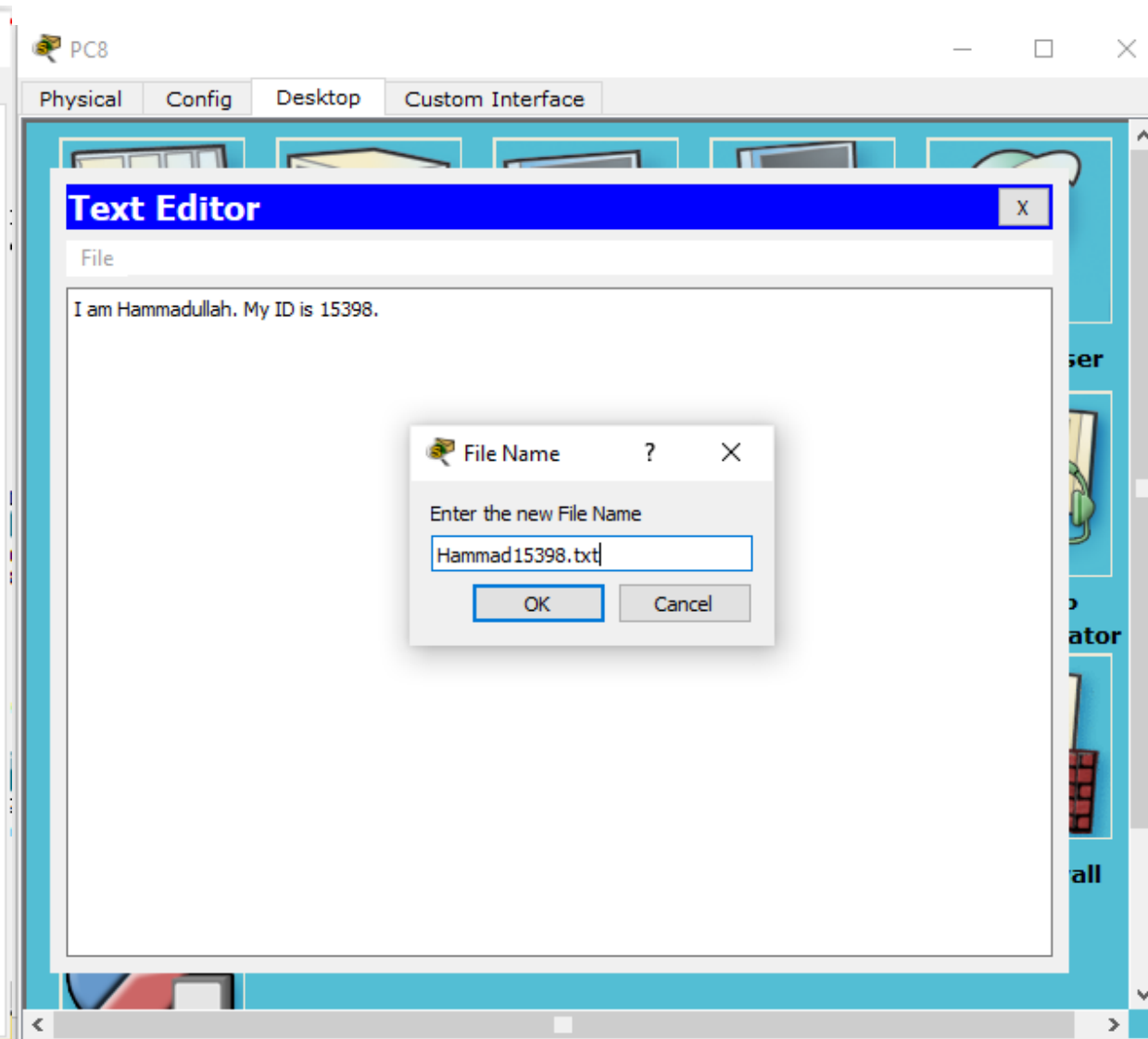
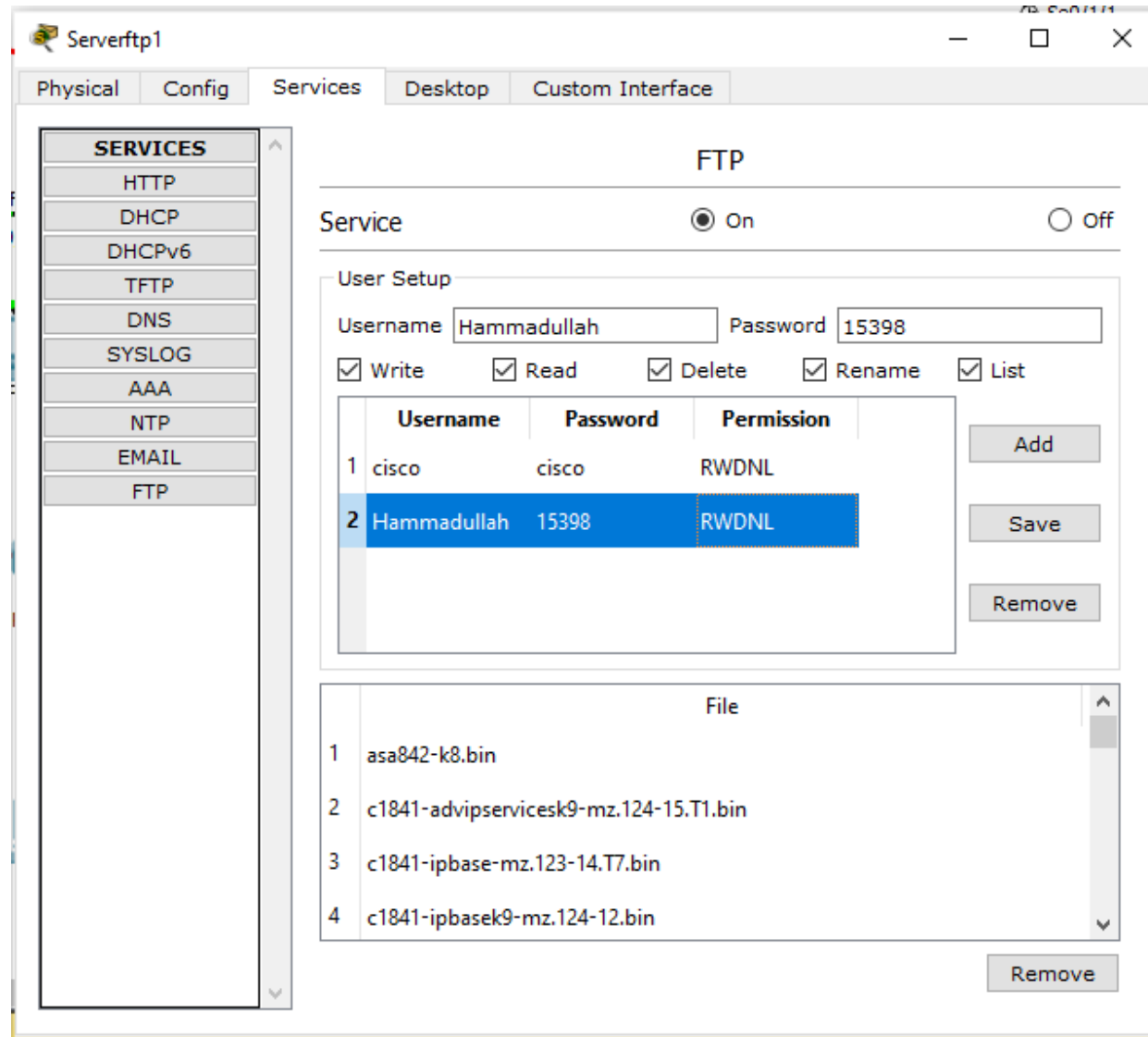
Link Local Address FE80::201:C9FF:FE9D:DE10

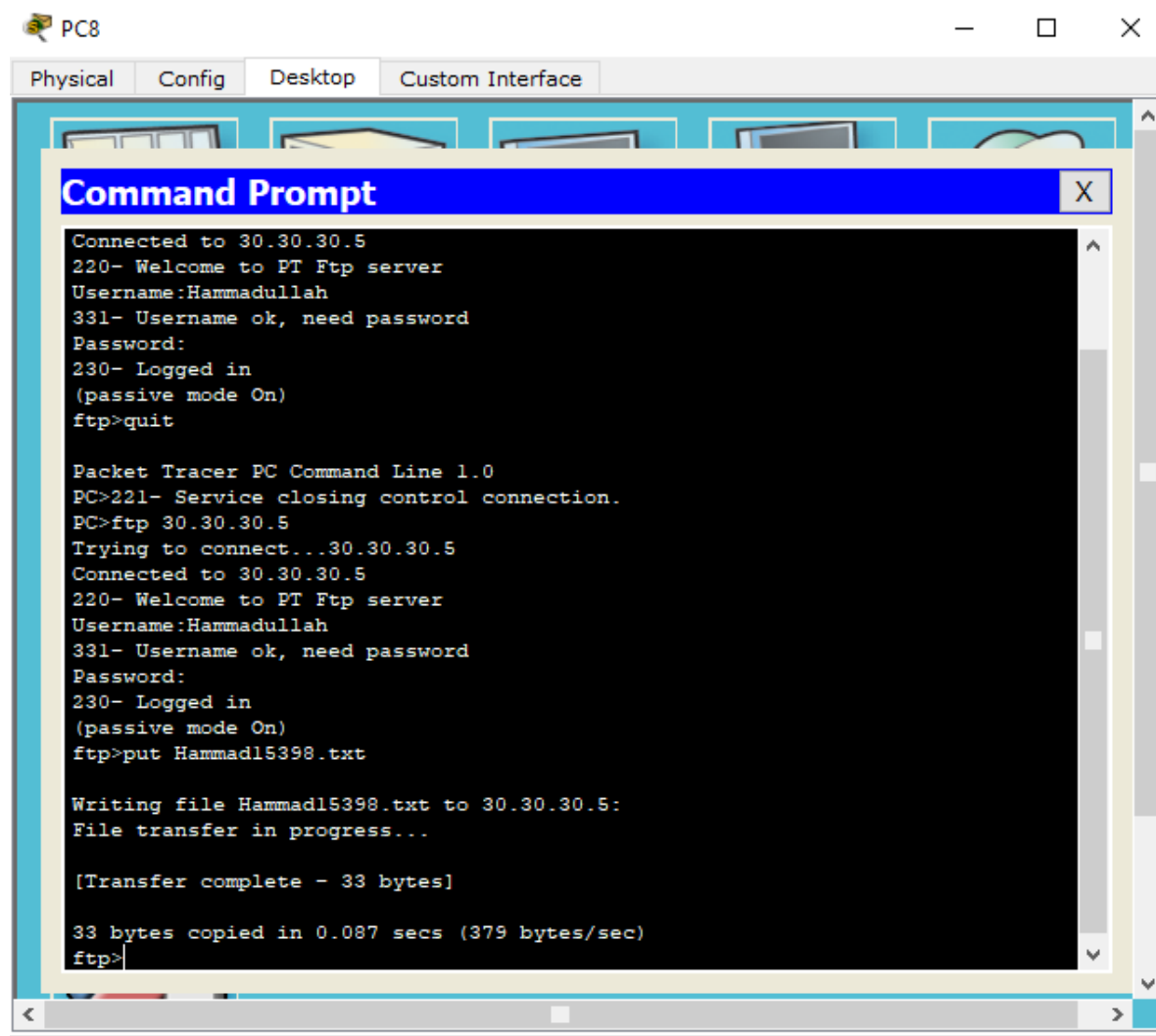
IPv6 Gateway

IPv6 DNS Server

FTP

Router 4





PC8

Physical Config Desktop Custom Interface

Command Prompt

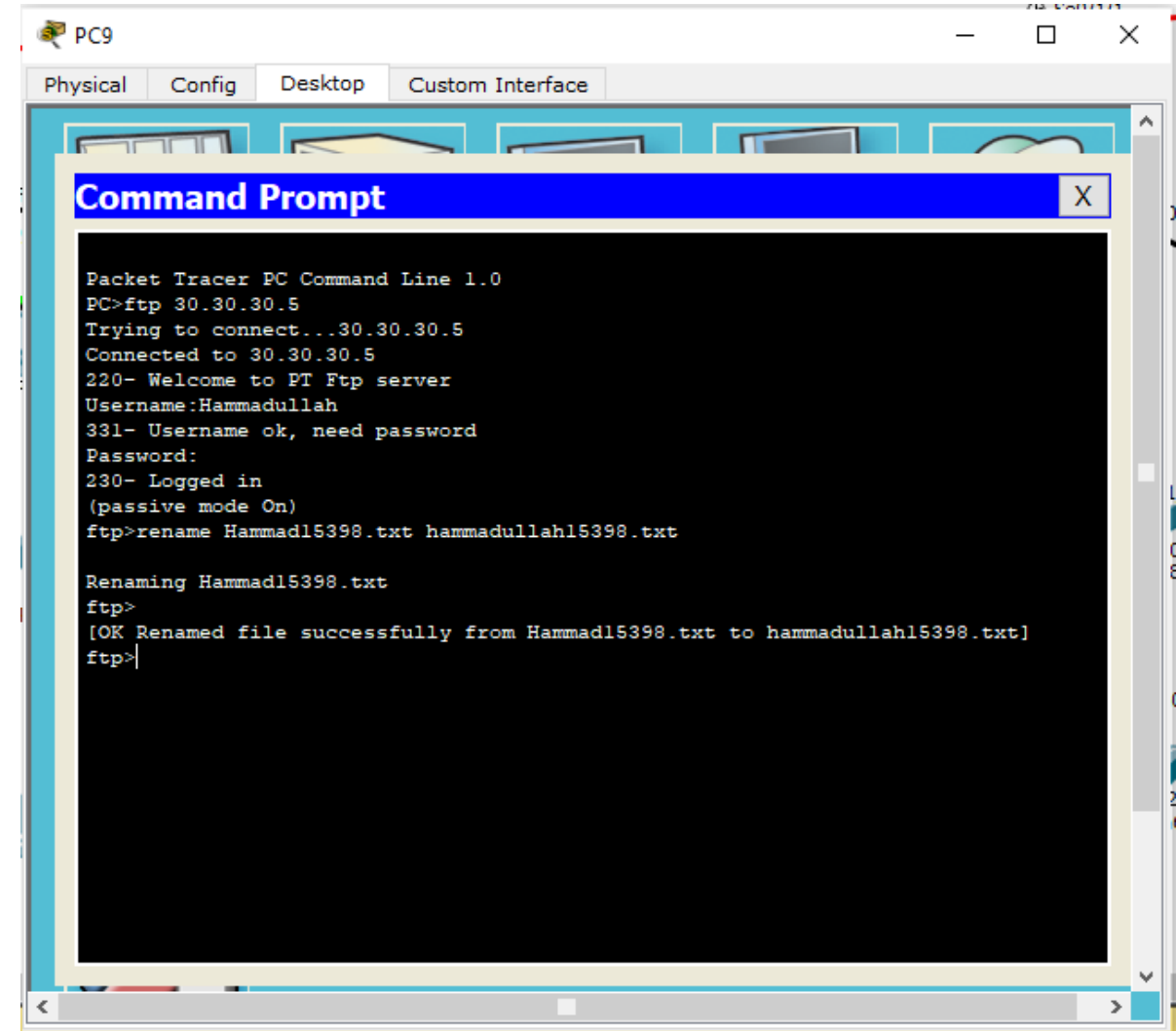
```
Connected to 30.30.30.5
220- Welcome to PT Ftp server
Username:Hammadullah
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>quit

Packet Tracer PC Command Line 1.0
PC>221- Service closing control connection.
PC>ftp 30.30.30.5
Trying to connect...30.30.30.5
Connected to 30.30.30.5
220- Welcome to PT Ftp server
Username:Hammadullah
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>put Hammad15398.txt

Writing file Hammad15398.txt to 30.30.30.5:
File transfer in progress...

[Transfer complete - 33 bytes]

33 bytes copied in 0.087 secs (379 bytes/sec)
ftp>
```



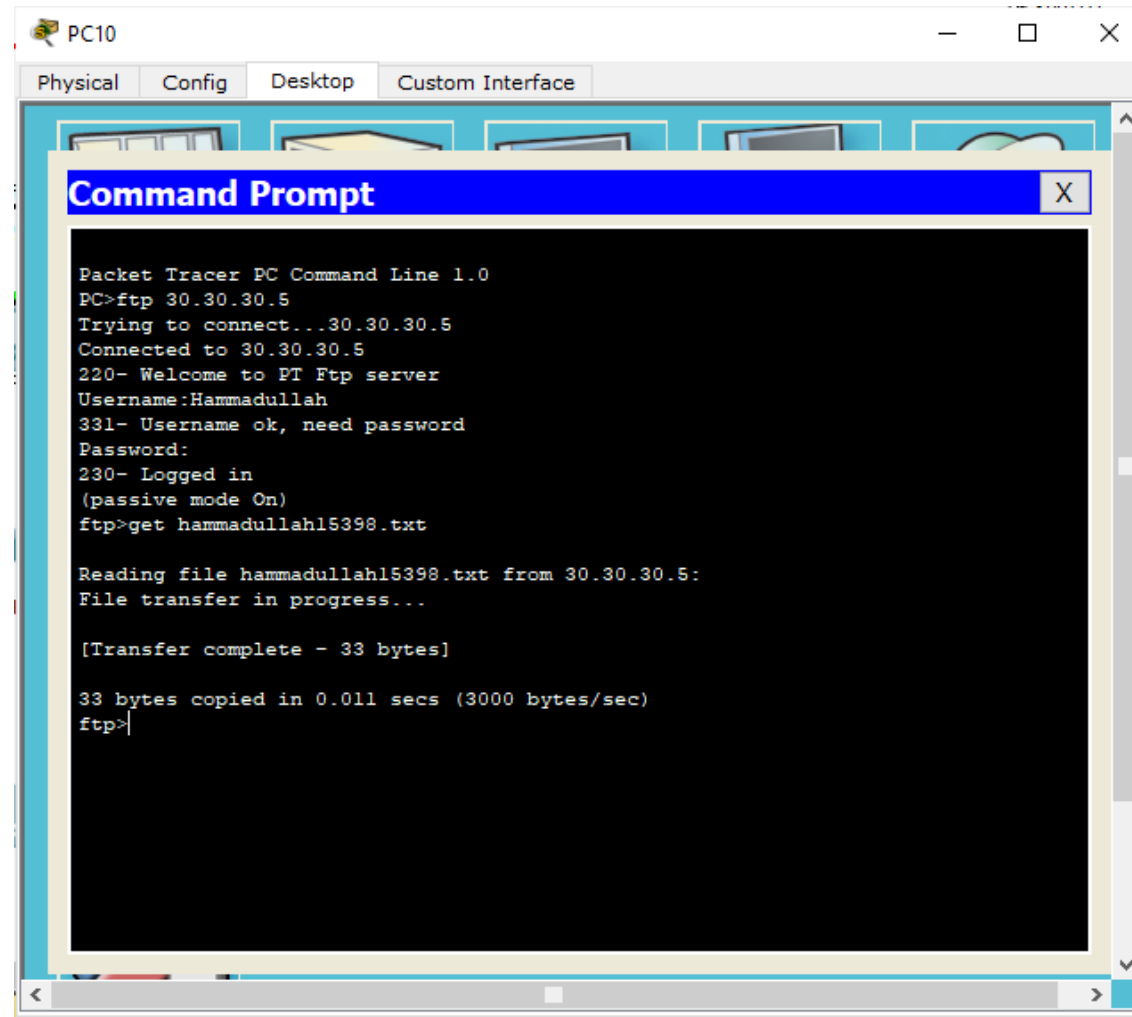
PC9

Physical Config Desktop Custom Interface

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ftp 30.30.30.5
Trying to connect...30.30.30.5
Connected to 30.30.30.5
220- Welcome to PT Ftp server
Username:Hammadullah
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>rename Hammad15398.txt hammadullah15398.txt

Renaming Hammad15398.txt
ftp>
[OK Renamed file successfully from Hammad15398.txt to hammadullah15398.txt]
ftp>
```



The screenshot shows a Packet Tracer PC window titled "PC10" with tabs for "Physical", "Config", "Desktop", and "Custom Interface". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt window has a blue title bar and a black background with white text. The text shows the execution of an FTP command to connect to 30.30.30.5, log in with the username "Hammadullah", and retrieve a file named "hammadullah15398.txt". The transfer is successful, with 33 bytes copied in 0.011 seconds.

```
Packet Tracer PC Command Line 1.0
PC>ftp 30.30.30.5
Trying to connect...30.30.30.5
Connected to 30.30.30.5
220- Welcome to PT Ftp server
Username:Hammadullah
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>get hammadullah15398.txt

Reading file hammadullah15398.txt from 30.30.30.5:
File transfer in progress...

[Transfer complete - 33 bytes]

33 bytes copied in 0.011 secs (3000 bytes/sec)
ftp>
```

Router 5

The screenshot shows the 'Serverftp2' configuration window with the 'Services' tab selected. The 'FTP' service is turned 'On'. Under 'User Setup', the 'Hammad' user is configured with password '15398' and 'RWDNL' permissions. A table lists existing users, and a 'File' list shows various system files.

Services

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP

FTP

Service ☒ On ☐ Off

User Setup

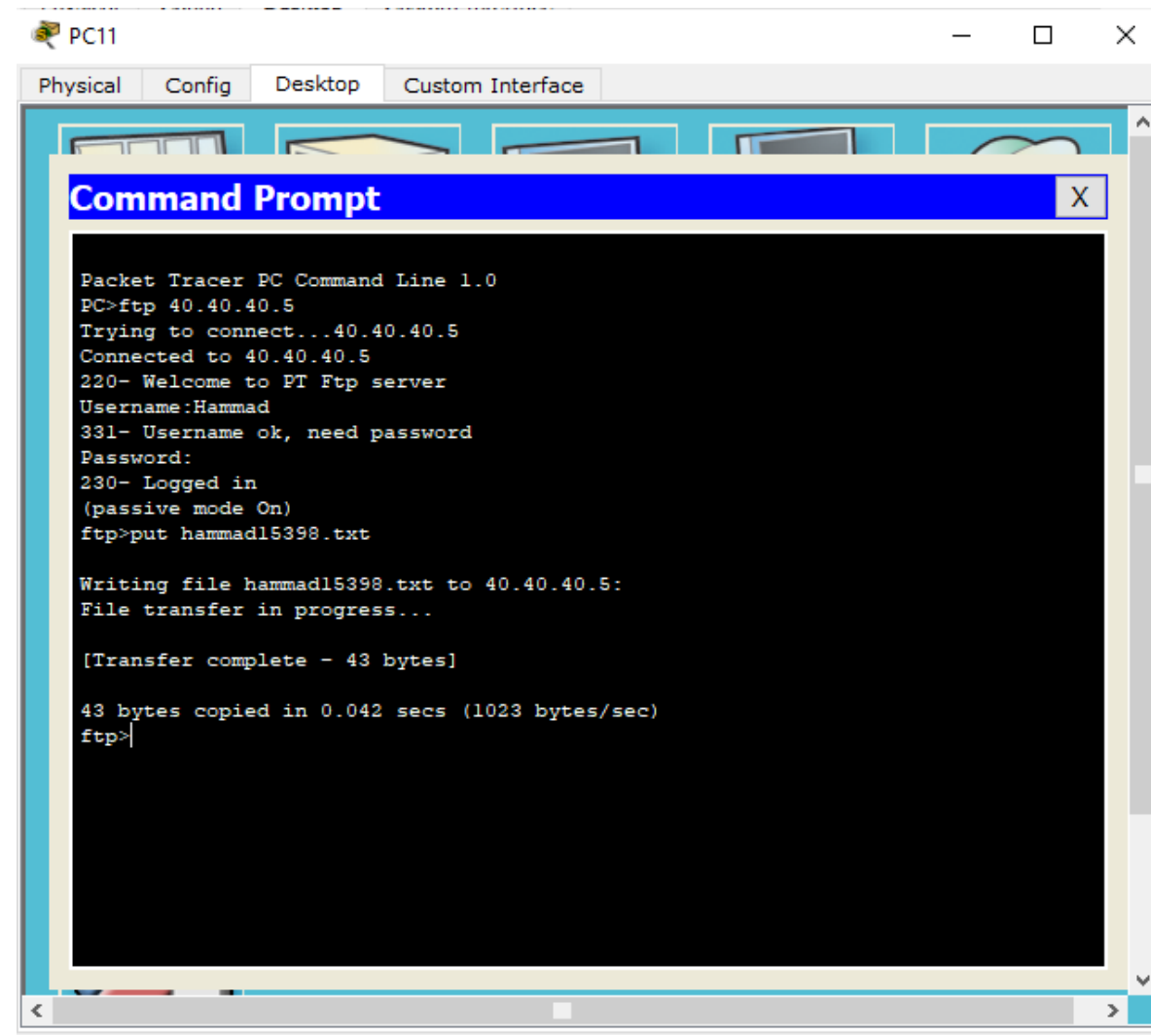
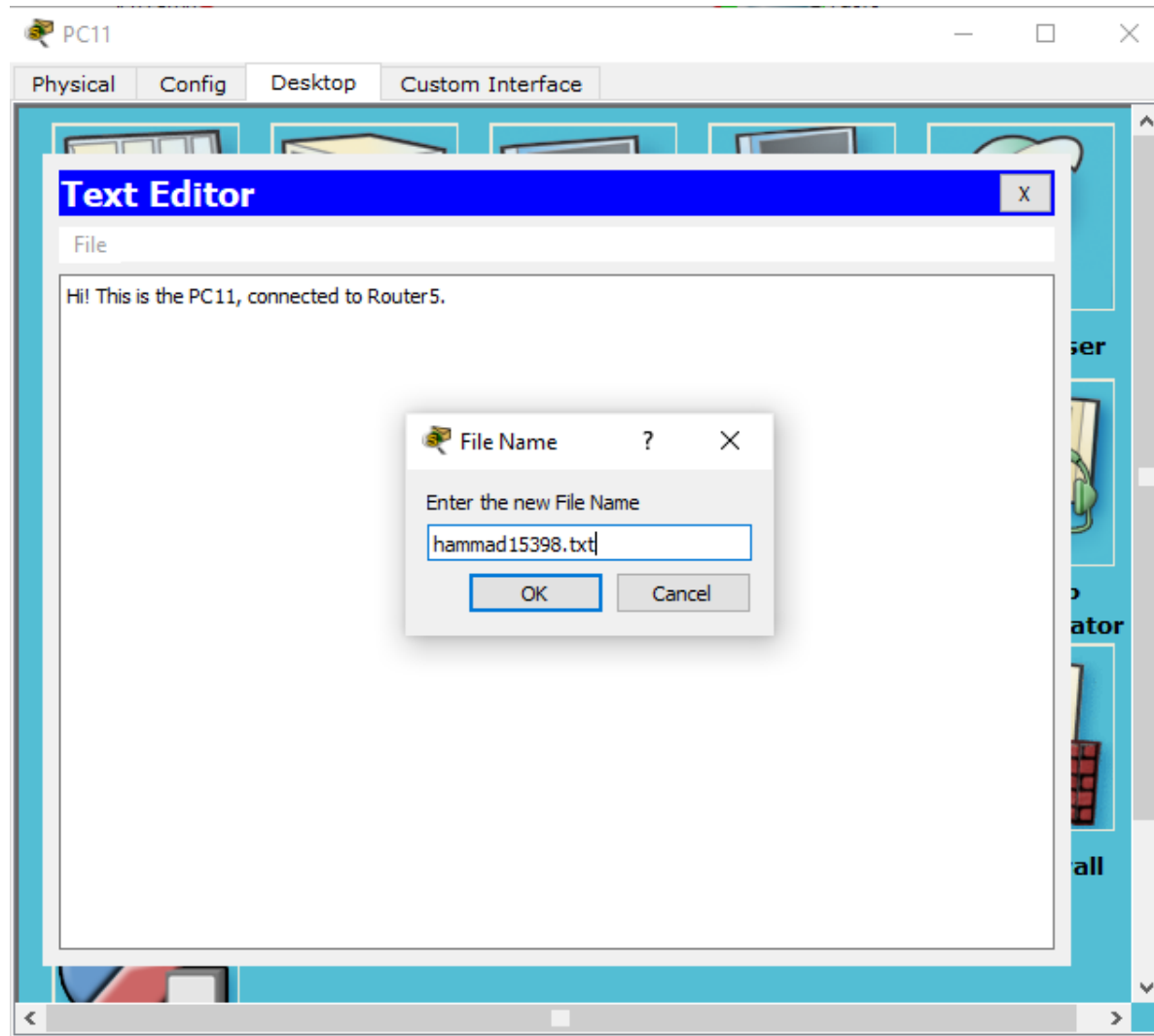
Username Password

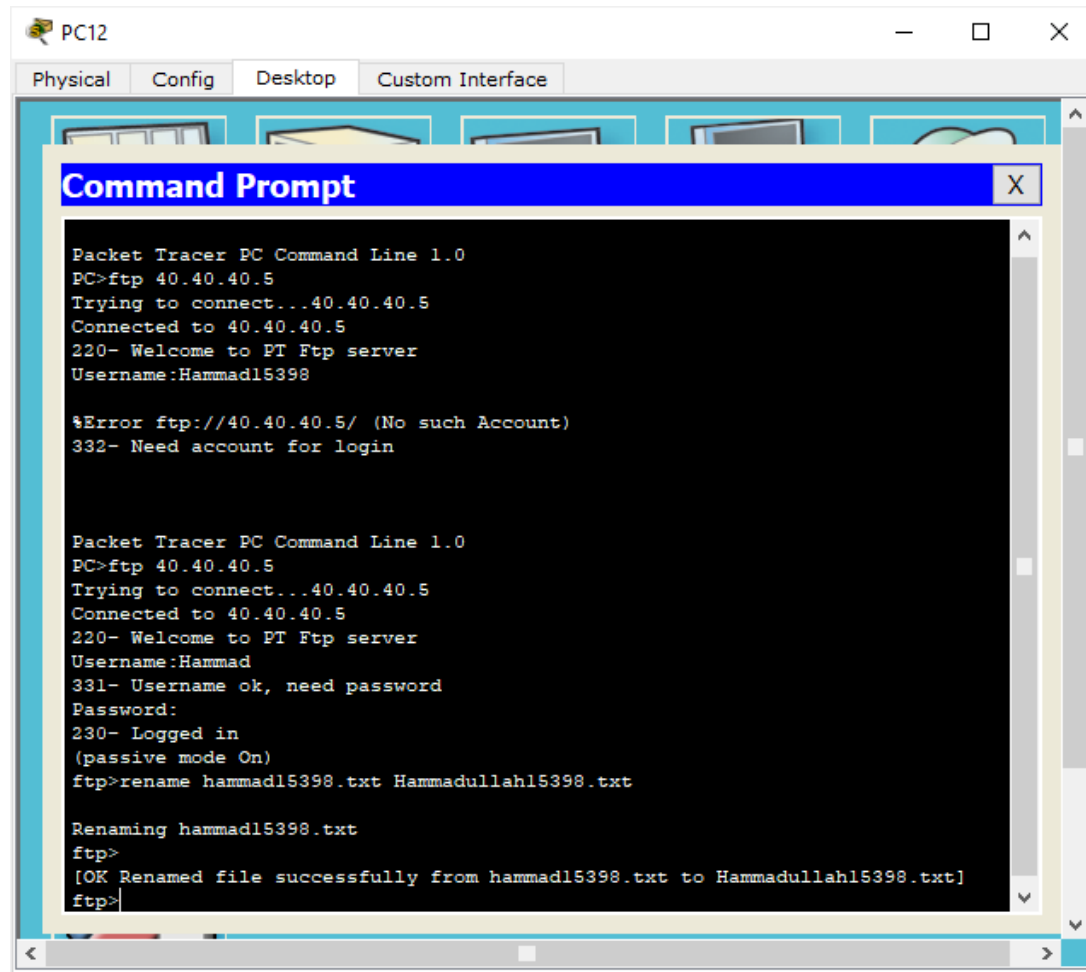
☒ Write ☒ Read ☒ Delete ☒ Rename ☒ List

	Username	Password	Permission
1	cisco	cisco	RWDNL
2	Hammad	15398	RWDNL

File

1	asa842-k8.bin
2	c1841-advipservicesk9-mz.124-15.T1.bin
3	c1841-ipbase-mz.123-14.T7.bin
4	c1841-ipbasek9-mz.124-12.bin





PC12

Physical Config Desktop Custom Interface

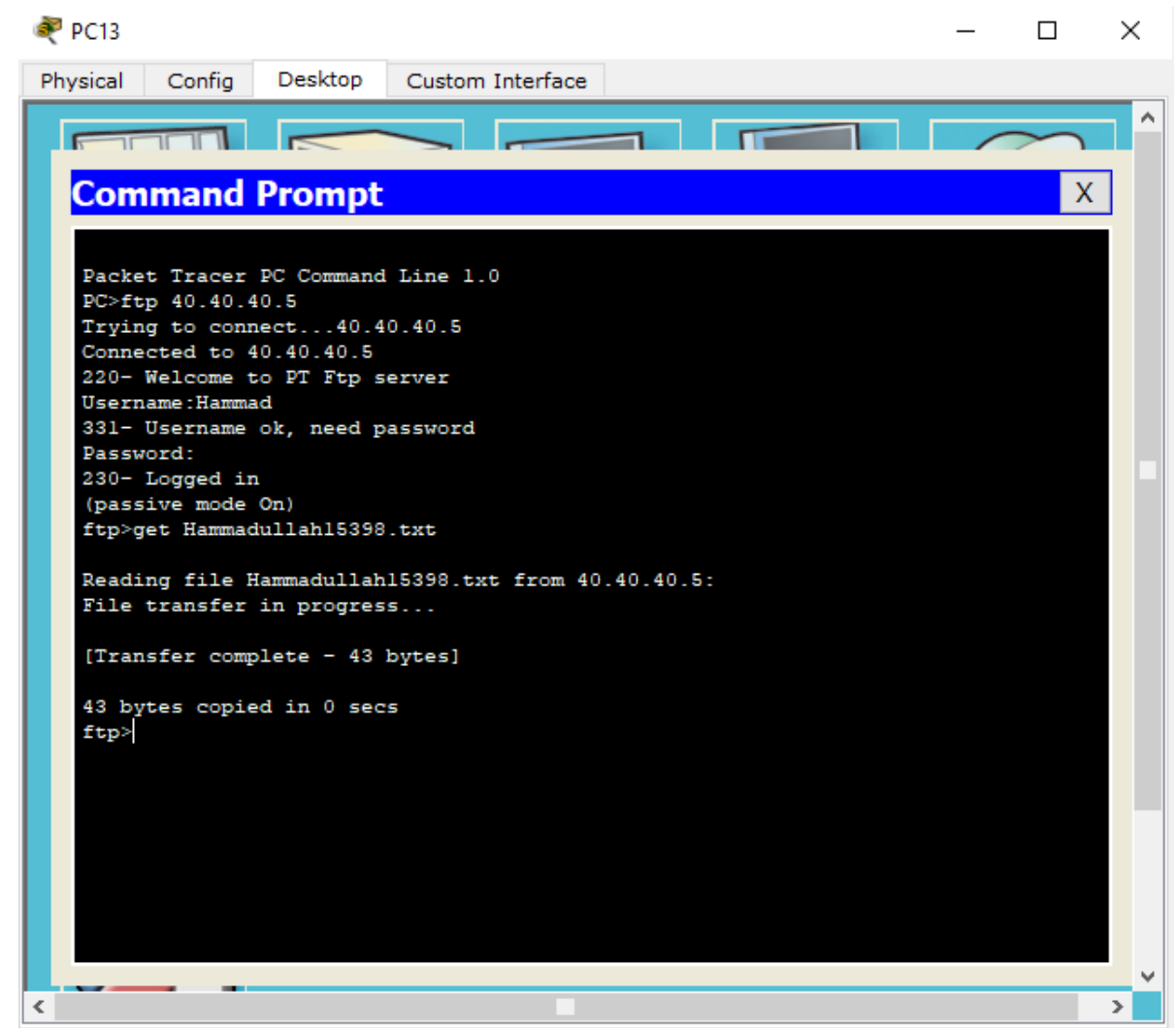
Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ftp 40.40.40.5
Trying to connect...40.40.40.5
Connected to 40.40.40.5
220- Welcome to PT Ftp server
Username:Hammad15398

%Error ftp://40.40.40.5/ (No such Account)
332- Need account for login

Packet Tracer PC Command Line 1.0
PC>ftp 40.40.40.5
Trying to connect...40.40.40.5
Connected to 40.40.40.5
220- Welcome to PT Ftp server
Username:Hammad
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>rename hammad15398.txt Hammadullah15398.txt

Renaming hammad15398.txt
ftp>
[OK Renamed file successfully from hammad15398.txt to Hammadullah15398.txt]
ftp>
```



PC13

Physical Config Desktop Custom Interface

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ftp 40.40.40.5
Trying to connect...40.40.40.5
Connected to 40.40.40.5
220- Welcome to PT Ftp server
Username:Hammad
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>get Hammadullah15398.txt

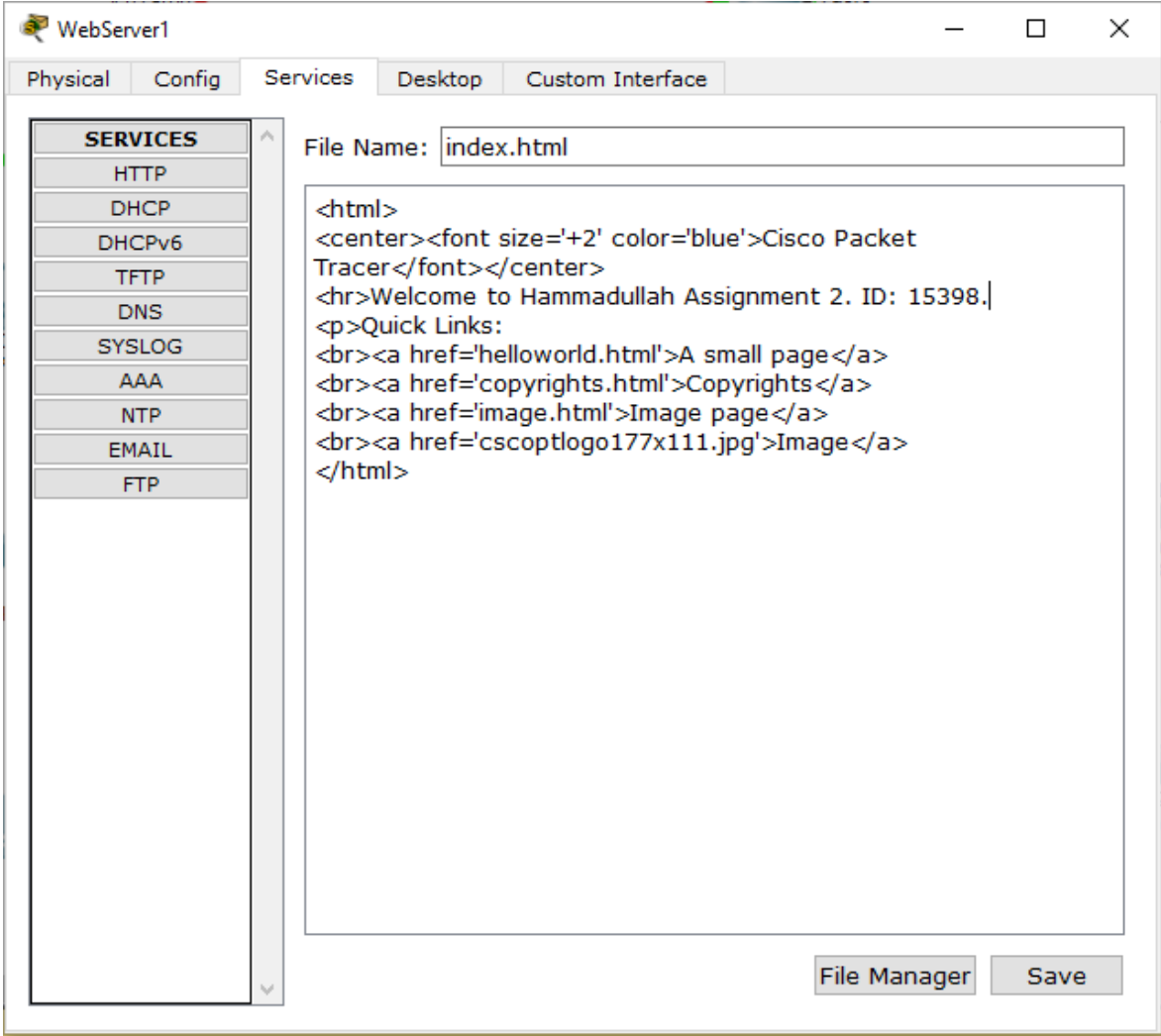
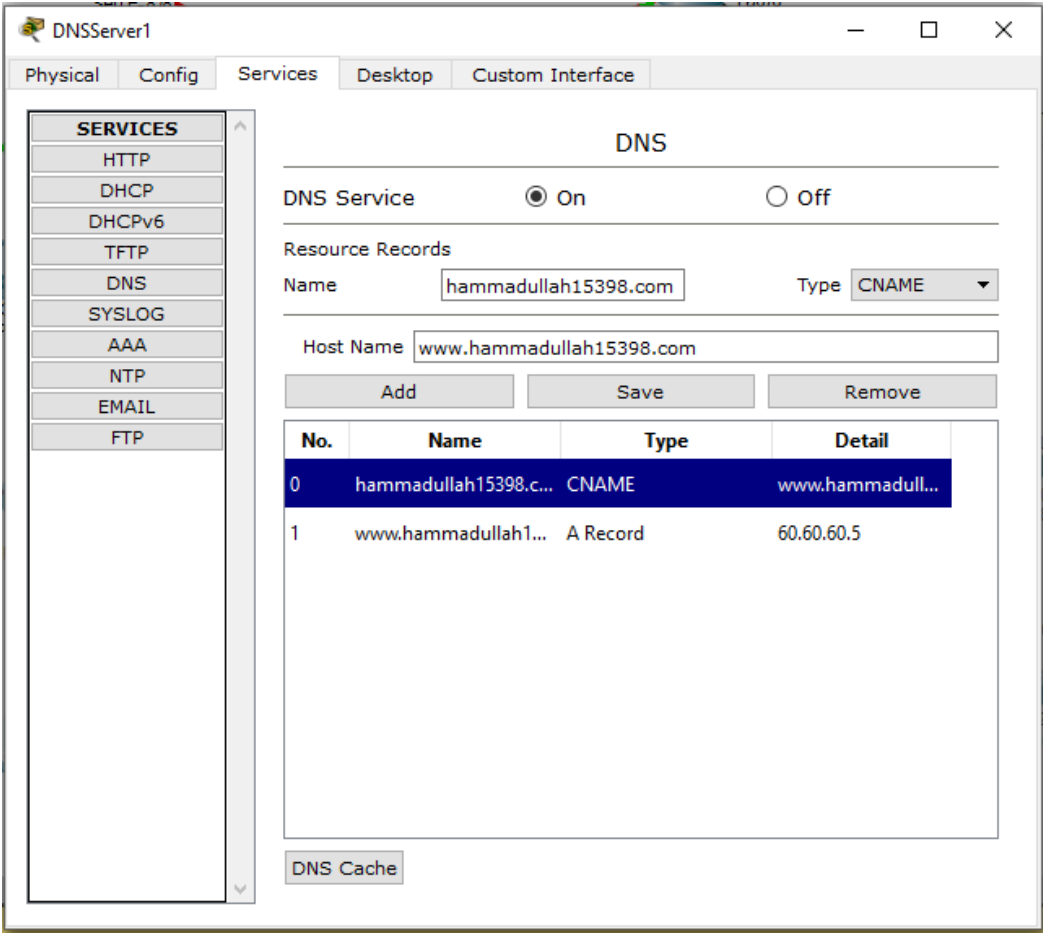
Reading file Hammadullah15398.txt from 40.40.40.5:
File transfer in progress...

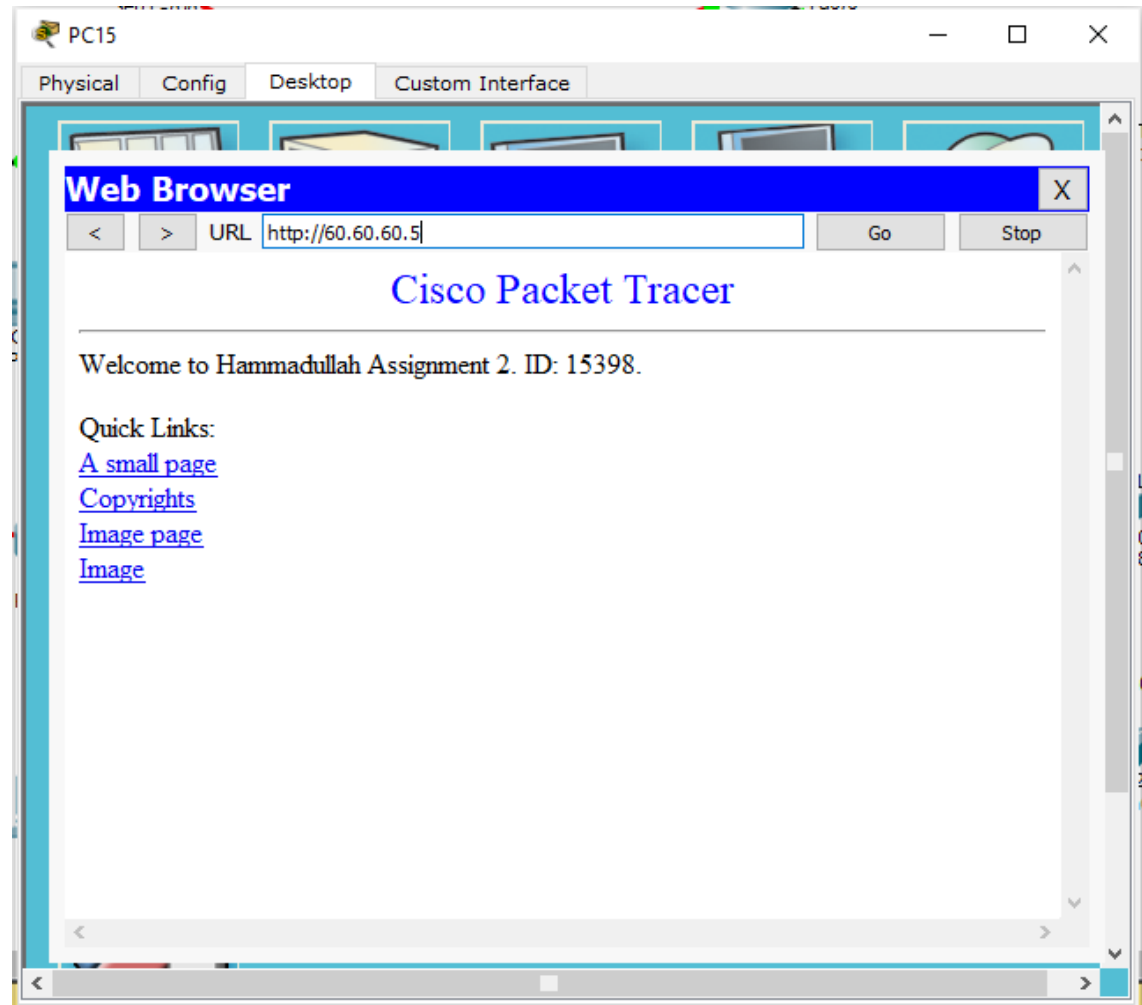
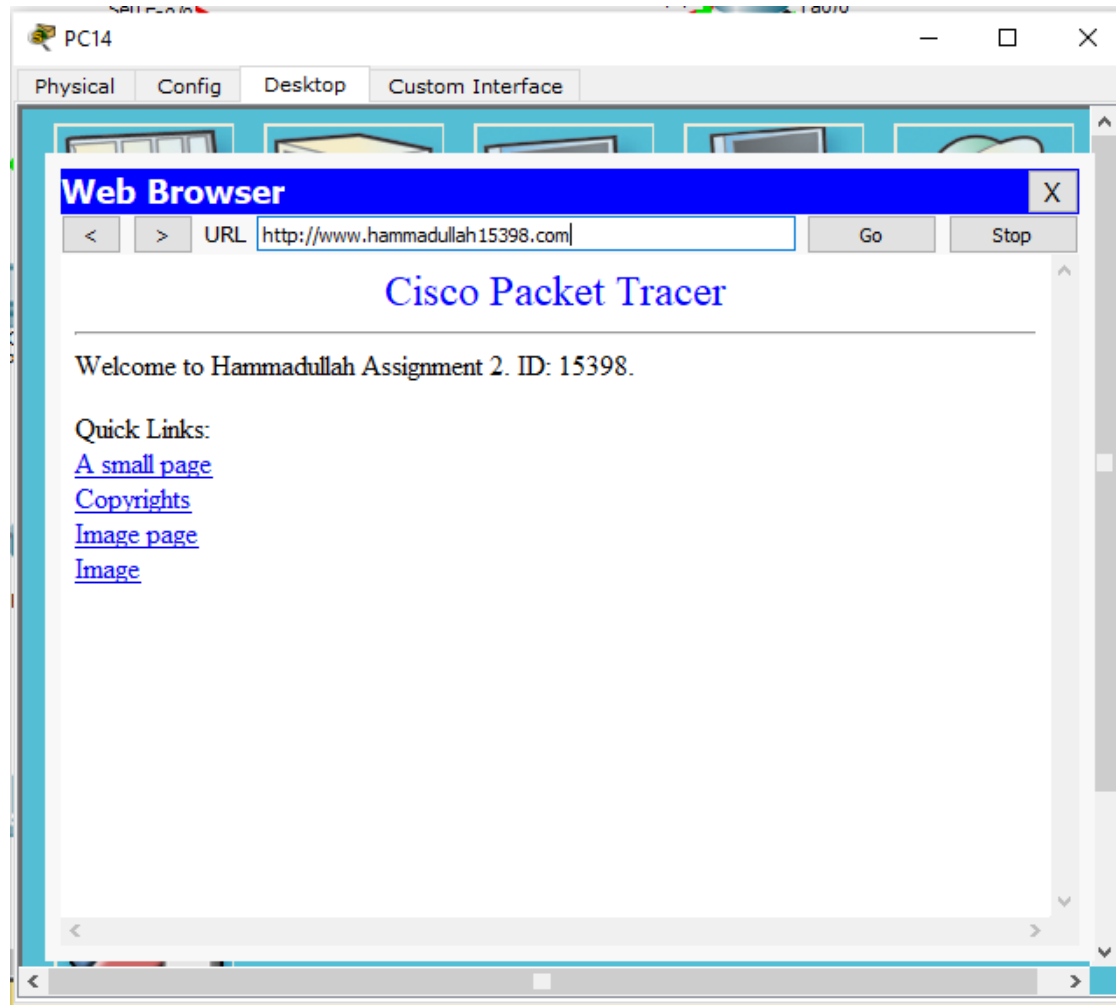
[Transfer complete - 43 bytes]

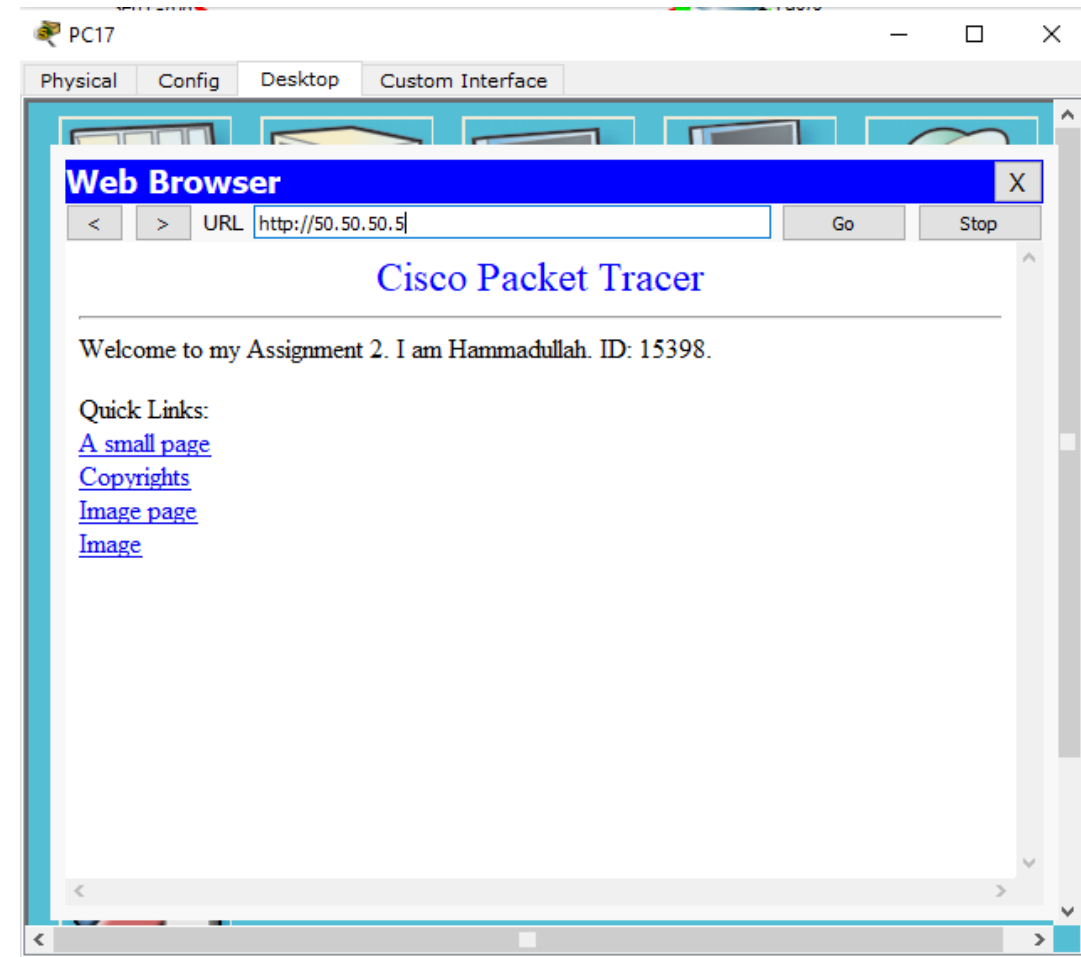
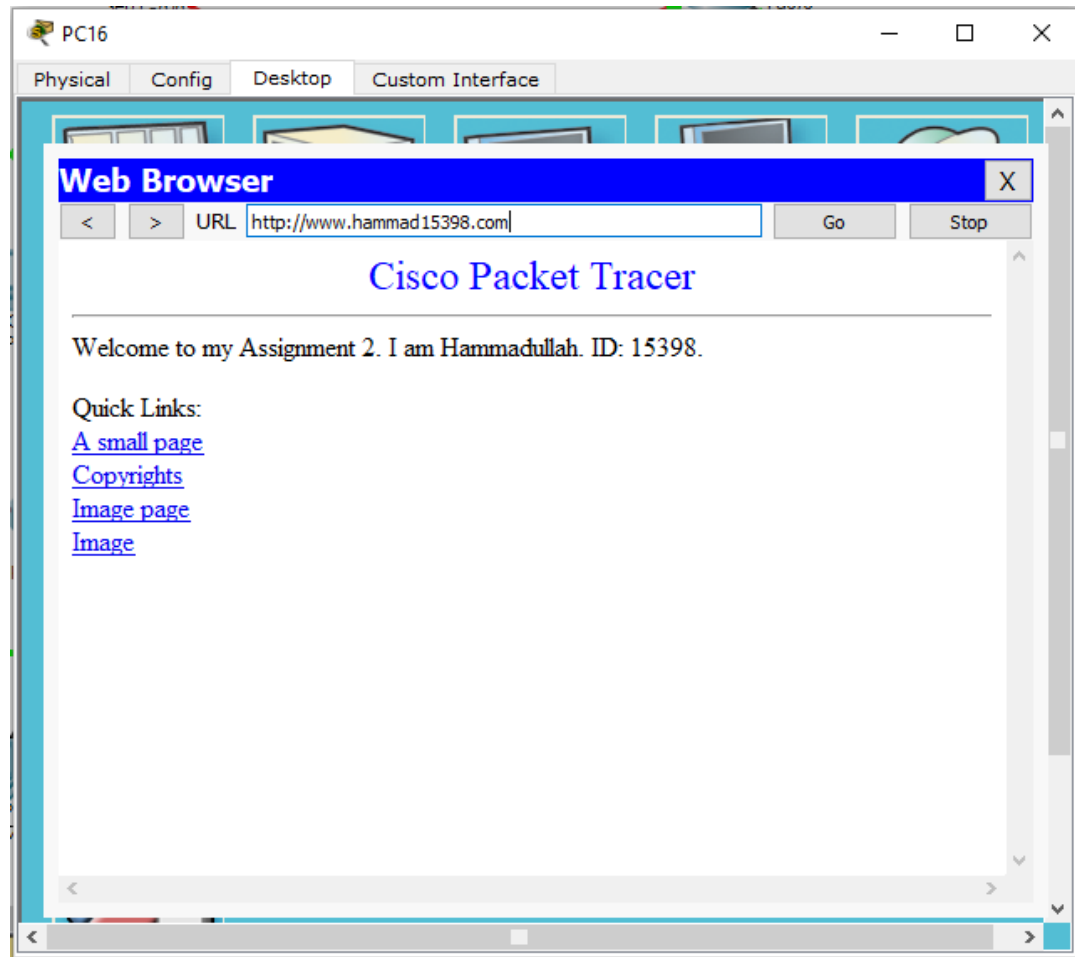
43 bytes copied in 0 secs
ftp>
```

Web Server

Router 7







Email Server

Router 8(Here according to in switch there will one extra pc bcz we need to send email to one another which is not in the topology, so I add there a pc bcz we have DNS and email server in both routers 8 and 9.so no need to routing and just email is sent from one another.)

The screenshot shows the 'EmailServer1' configuration window with the 'Services' tab selected. On the left, a list of services includes HTTP, DHCP, DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL, and FTP. The 'EMAIL' service is highlighted. The main area is titled 'EMAIL' and contains the following configuration options:

- SMTP Service:** A group box containing two radio buttons: 'ON' (selected) and 'OFF'.
- POP3 Service:** A group box containing two radio buttons: 'ON' (selected) and 'OFF'.
- Domain Name:** A text field containing 'gmail.com' and a 'Set' button.
- User Setup:** A section containing:
 - User:** A text field with 'PC19'.
 - Password:** A text field with '1234'.
 - A list box containing 'PC19' (highlighted) and 'PC20'.
 - Buttons: '+', '-', 'Change', and 'Password'.

EmailServer1

Physical

Config

Services

Desktop

Custom Interface

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

EMAIL

SMTP Service

ON

OFF

POP3 Service

ON

OFF

Domain Name:

gmail.com

Set

User Setup

User

PC20

Password

15398

PC19

PC20

+

-

Change Password

PC19

Physical

Config

Desktop

Custom Interface

Configure Mail

User Information

Your Name:

PC19

Email Address

PC19@gmail.com

Server Information

Incoming Mail Server

gmail.com

Outgoing Mail Server

gmail.com

Logon Information

User Name:

PC19

Password:

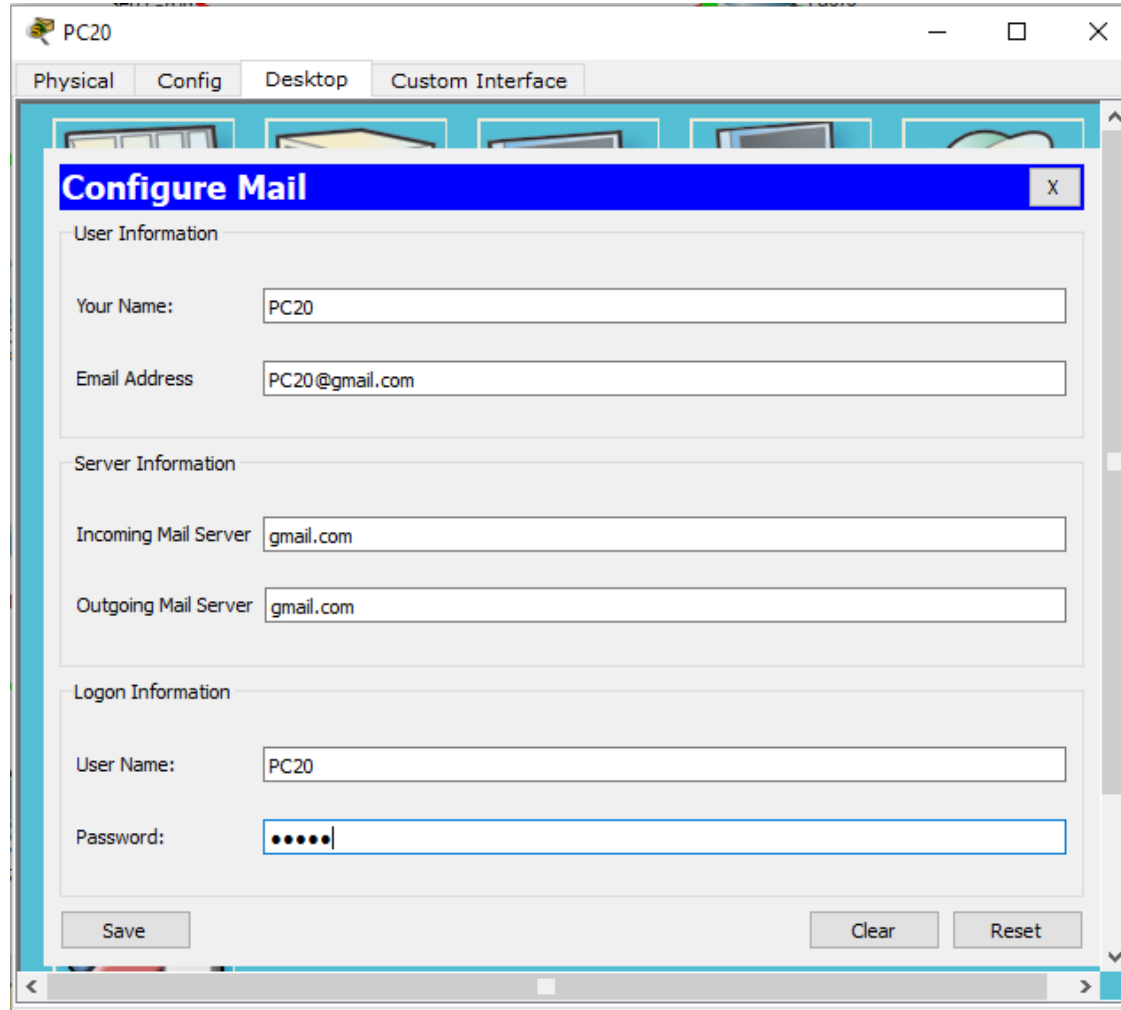
••••

Save

Clear

Reset

PC19 to PC 20



The screenshot shows the 'Configure Mail' dialog box on PC20. The dialog has three sections: User Information, Server Information, and Logon Information. The 'Your Name' field is filled with 'PC20'. The 'Email Address' field is filled with 'PC20@gmail.com'. The 'Incoming Mail Server' and 'Outgoing Mail Server' fields are both filled with 'gmail.com'. The 'User Name' field is filled with 'PC20'. The 'Password' field is filled with six dots. There are 'Save', 'Clear', and 'Reset' buttons at the bottom.

Configure Mail

User Information

Your Name: PC20

Email Address: PC20@gmail.com

Server Information

Incoming Mail Server: gmail.com

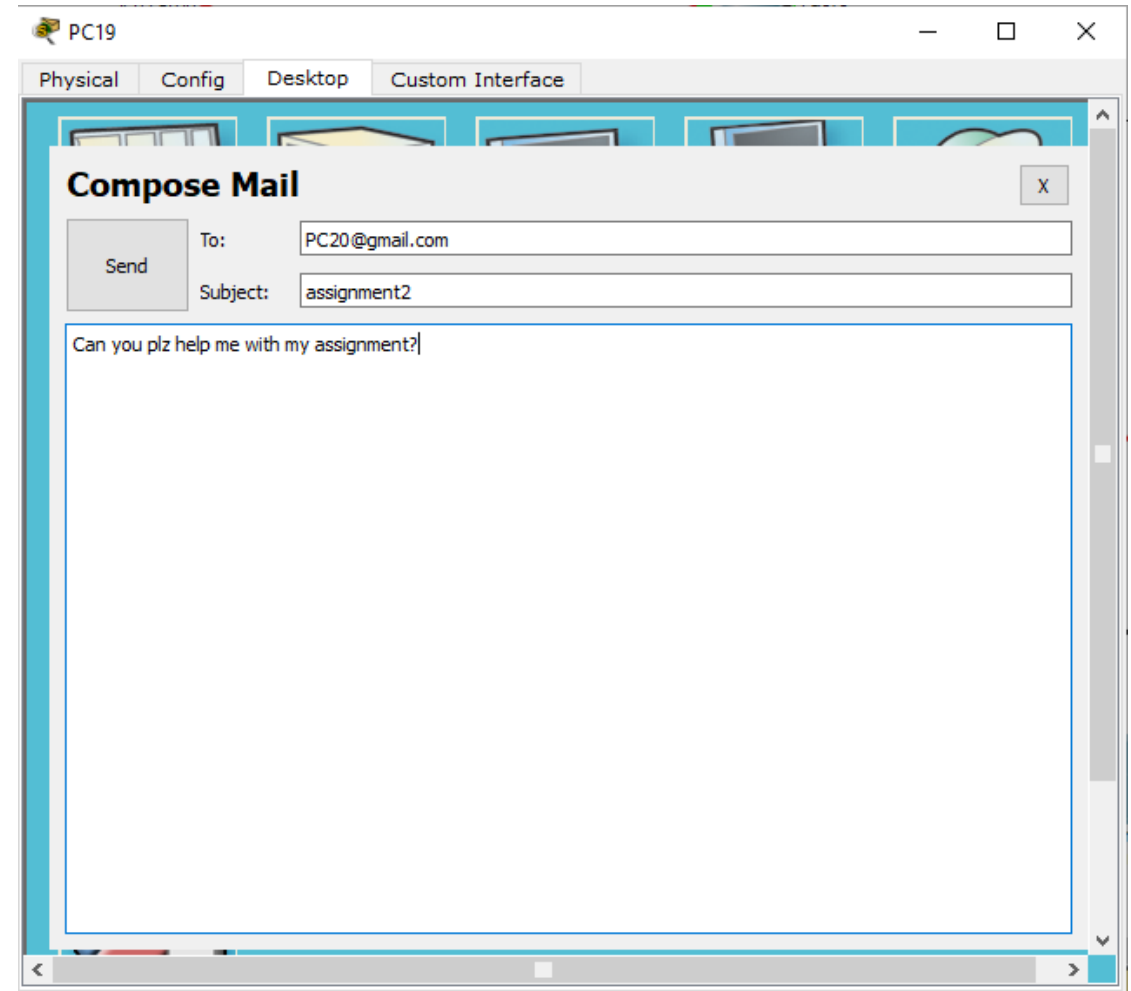
Outgoing Mail Server: gmail.com

Logon Information

User Name: PC20

Password: ••••••

Save Clear Reset



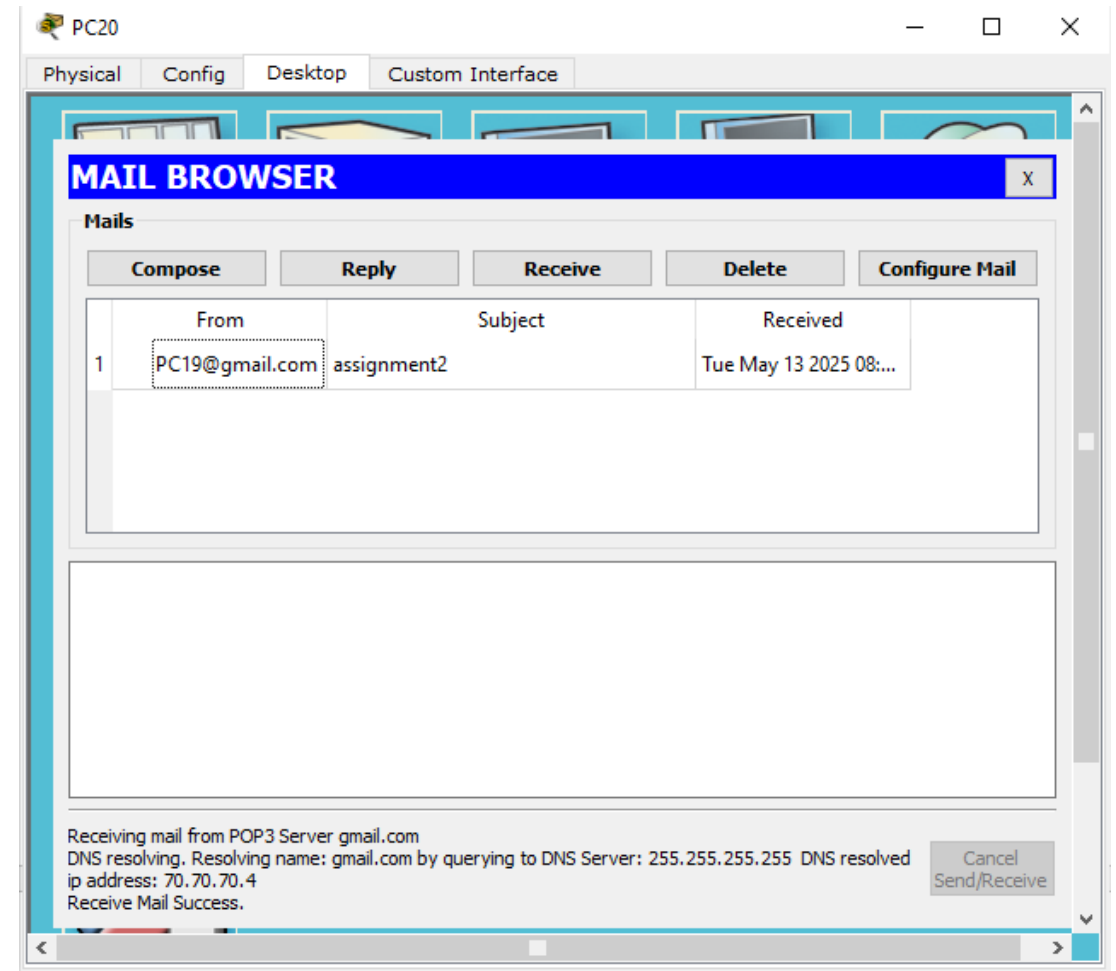
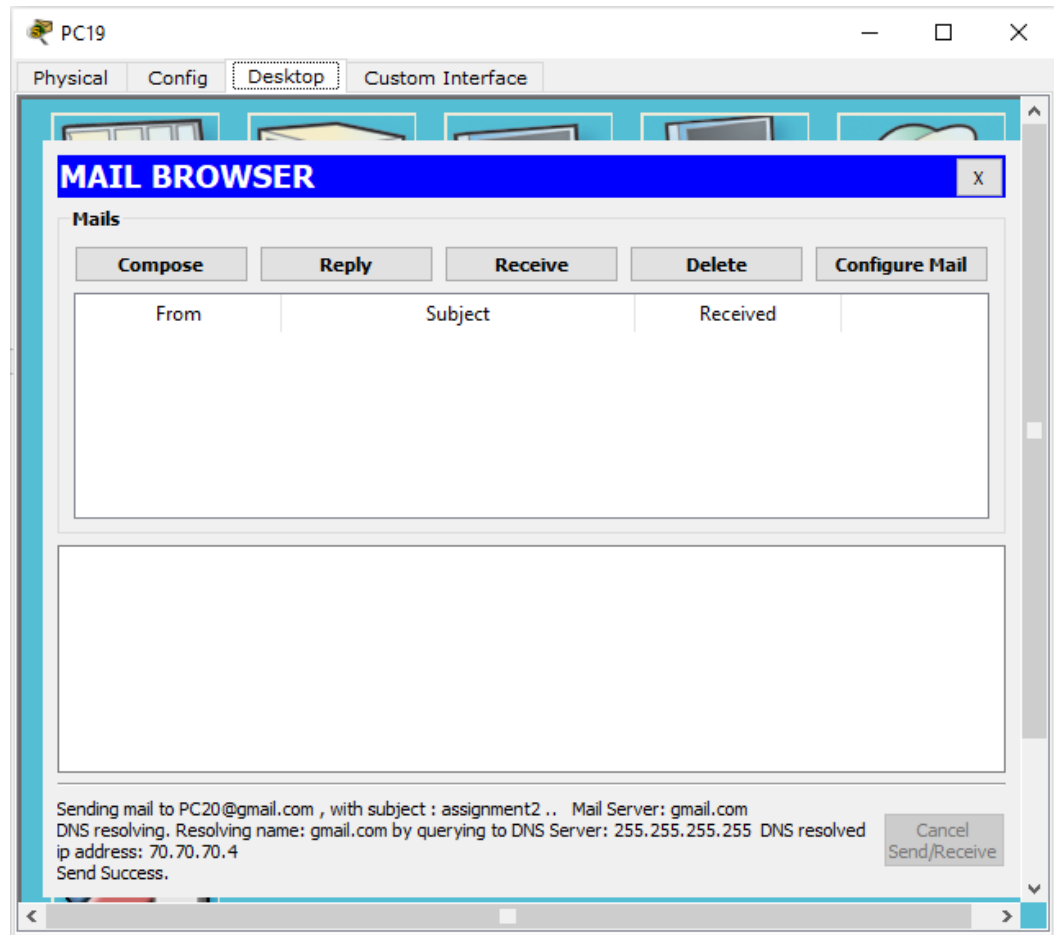
The screenshot shows the 'Compose Mail' dialog box on PC19. The 'To' field is filled with 'PC20@gmail.com'. The 'Subject' field is filled with 'assignment2'. The body of the email contains the text 'Can you plz help me with my assignment?'. There is a 'Send' button on the left.

Compose Mail

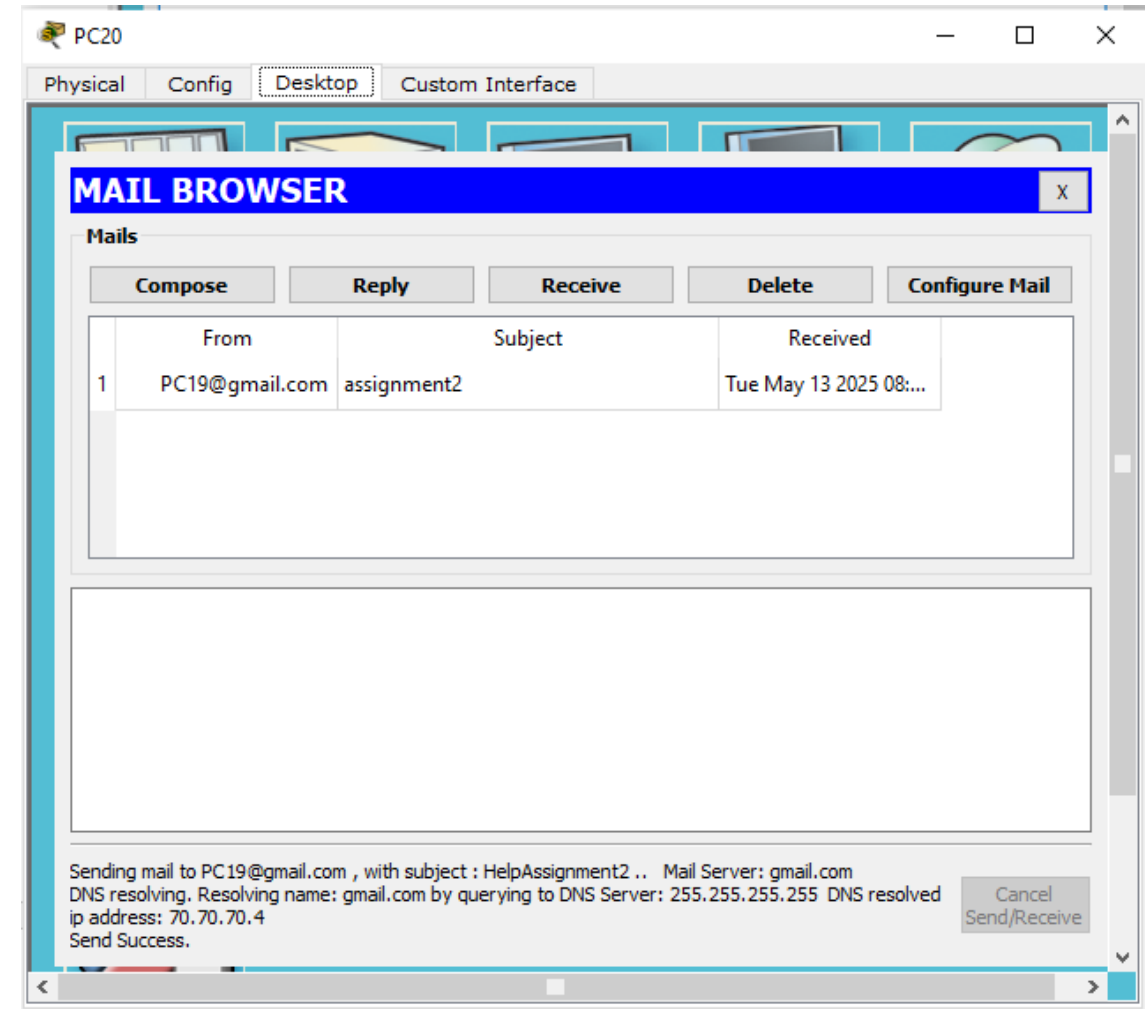
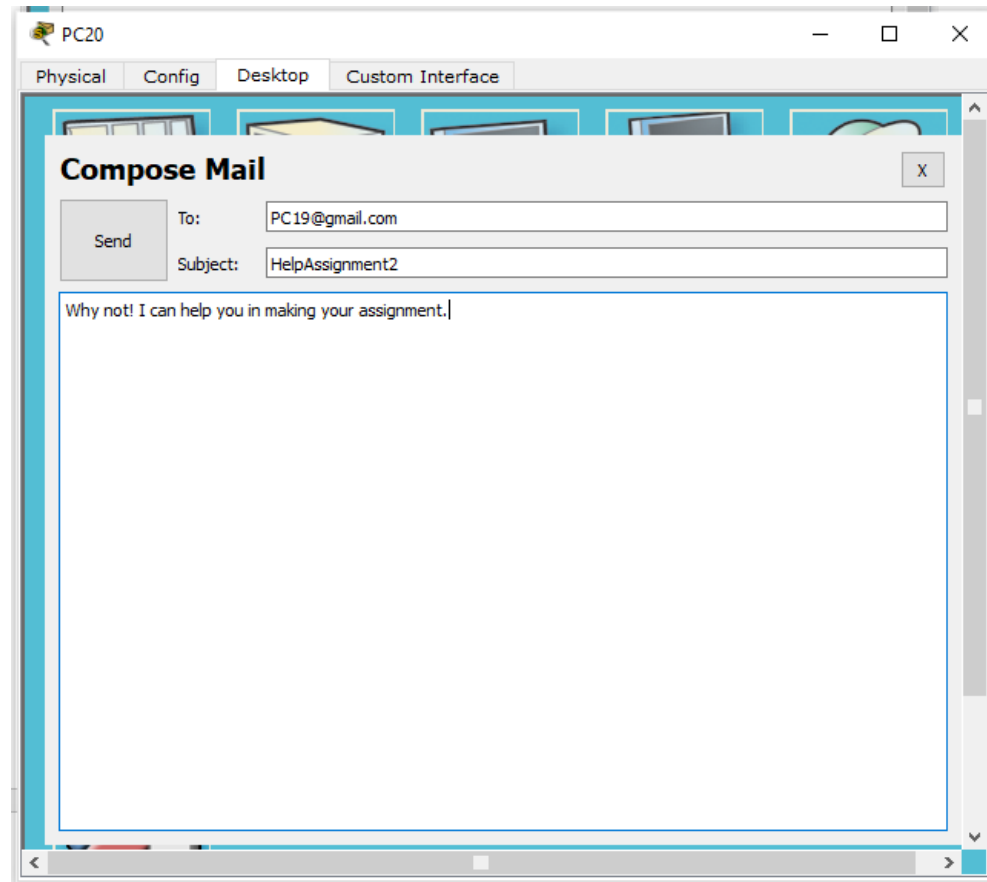
Send To: PC20@gmail.com

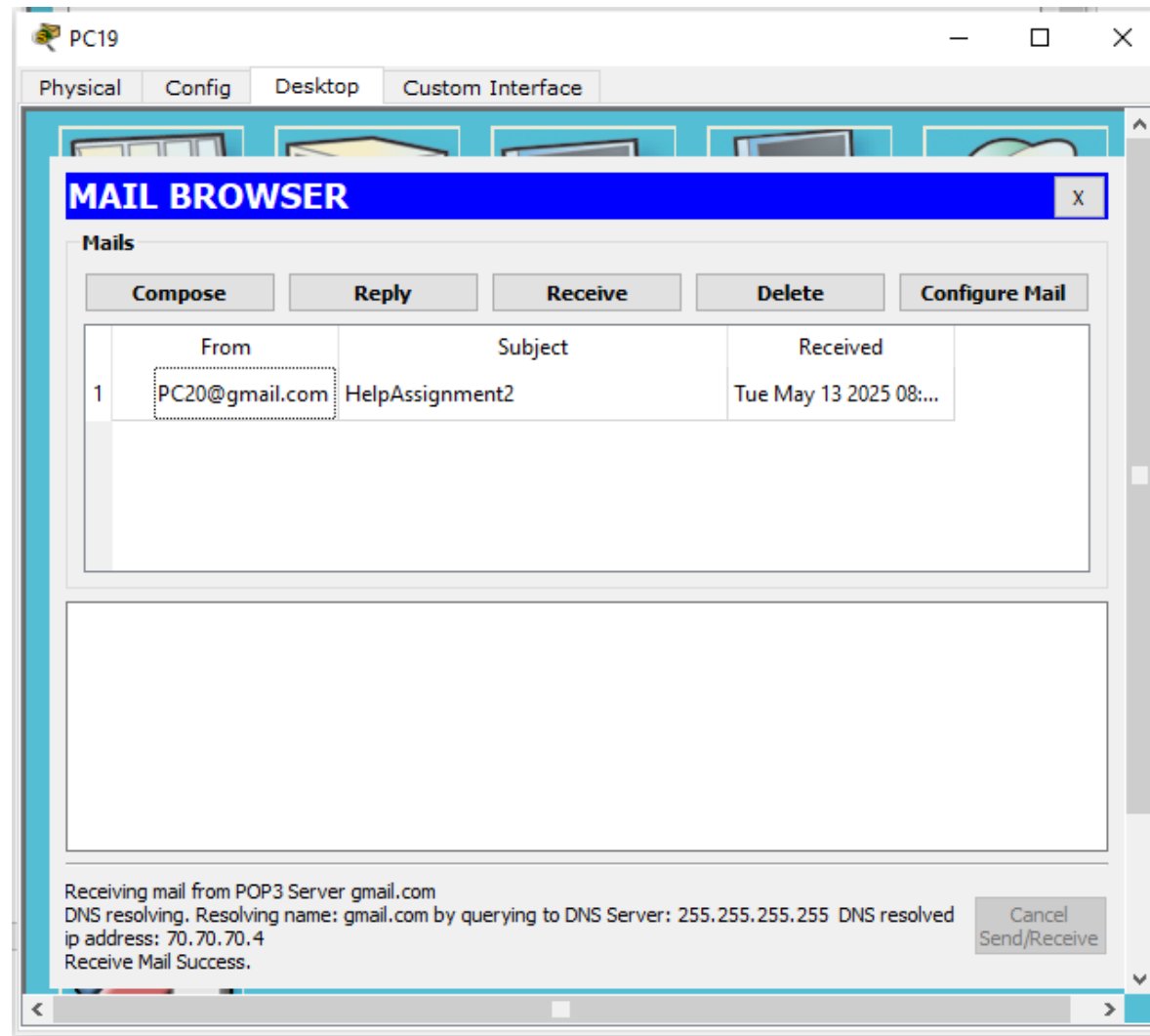
Subject: assignment2

Can you plz help me with my assignment?



Pc20 to PC19





Router 9

DNSServer4

PhysicalConfigServicesDesktopCustom Interface

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

DNS

DNS Service

On

Off

Resource Records

Name

gmail.com

Type

A Record

Address

80.80.80.4

Add

Save

Remove

No.	Name	Type	Detail
0	gmail.com	A Record	80.80.80.4

DNS Cache

EmailServer2

PhysicalConfigServicesDesktopCustom Interface

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

EMAIL

SMTP Service

ON

OFF

POP3 Service

ON

OFF

Domain Name:

gmail.com

Set

User Setup

User

PC21

Password

12345

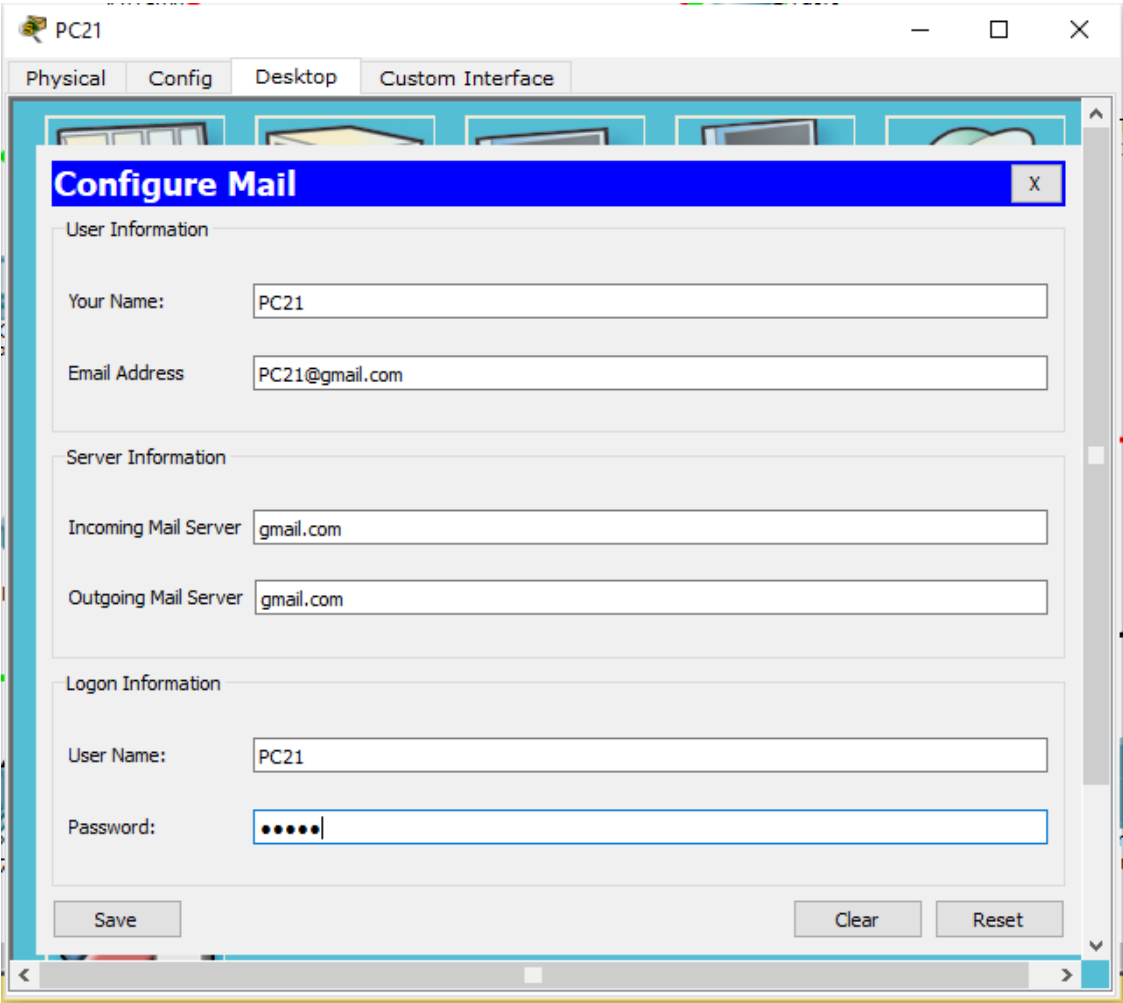
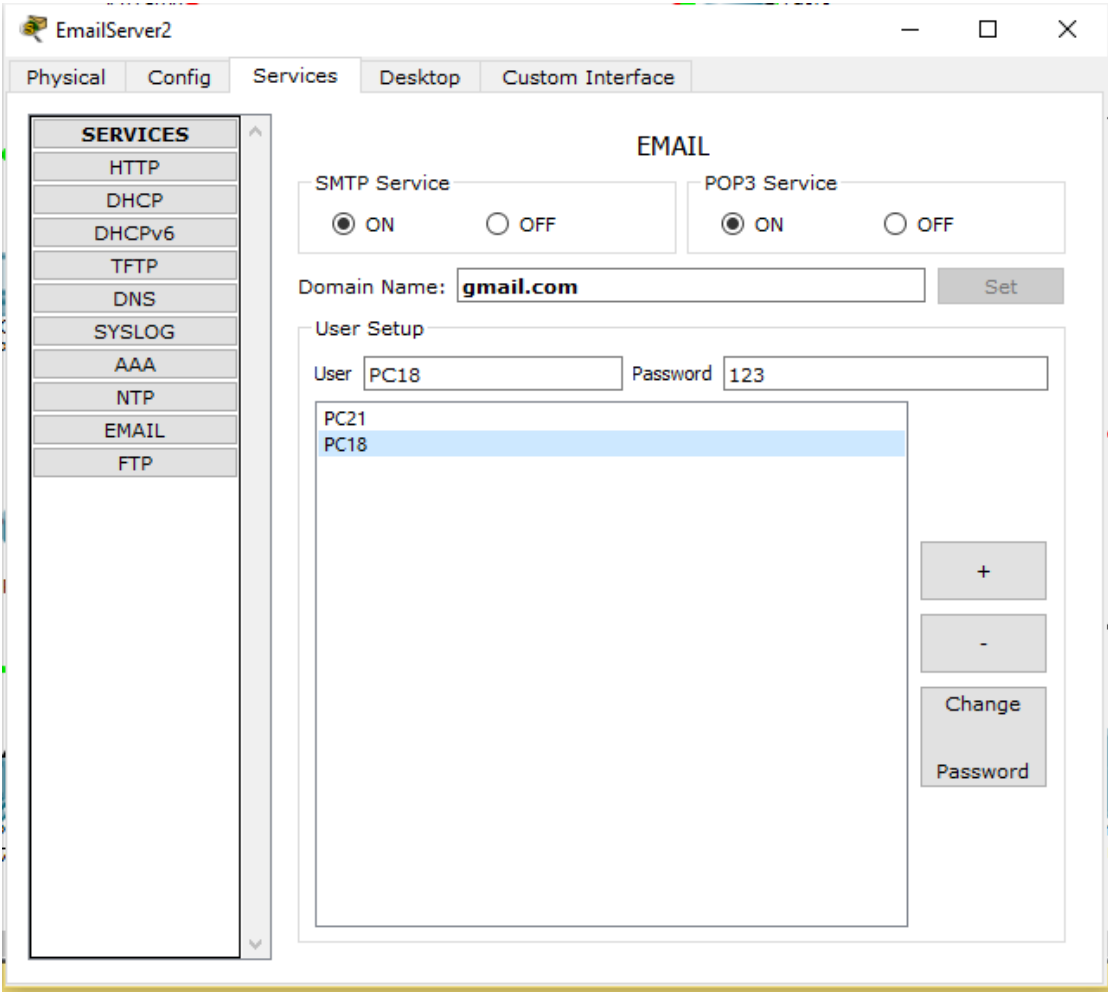
PC21

+

-

Change

Password



PC18

Physical Config Desktop Custom Interface

Configure Mail

User Information

Your Name: PC18

Email Address: PC18@gmail.com

Server Information

Incoming Mail Server: gmail.com

Outgoing Mail Server: gmail.com

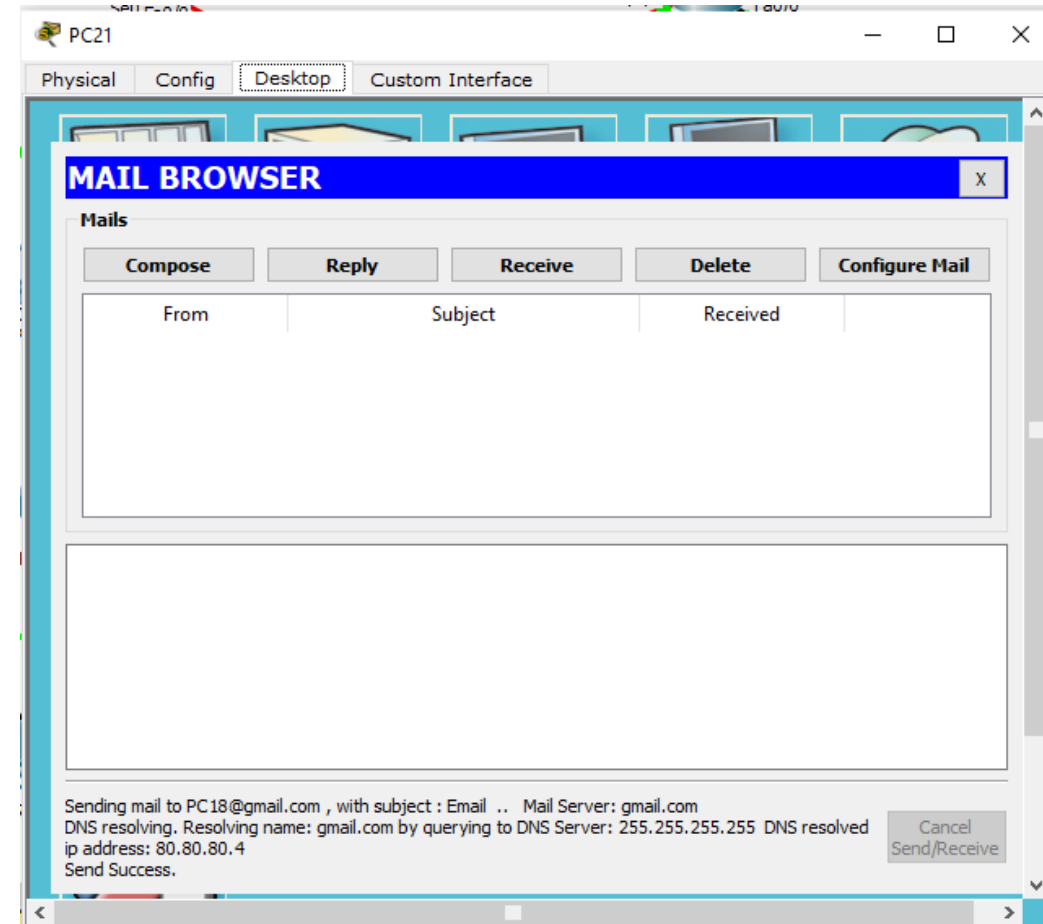
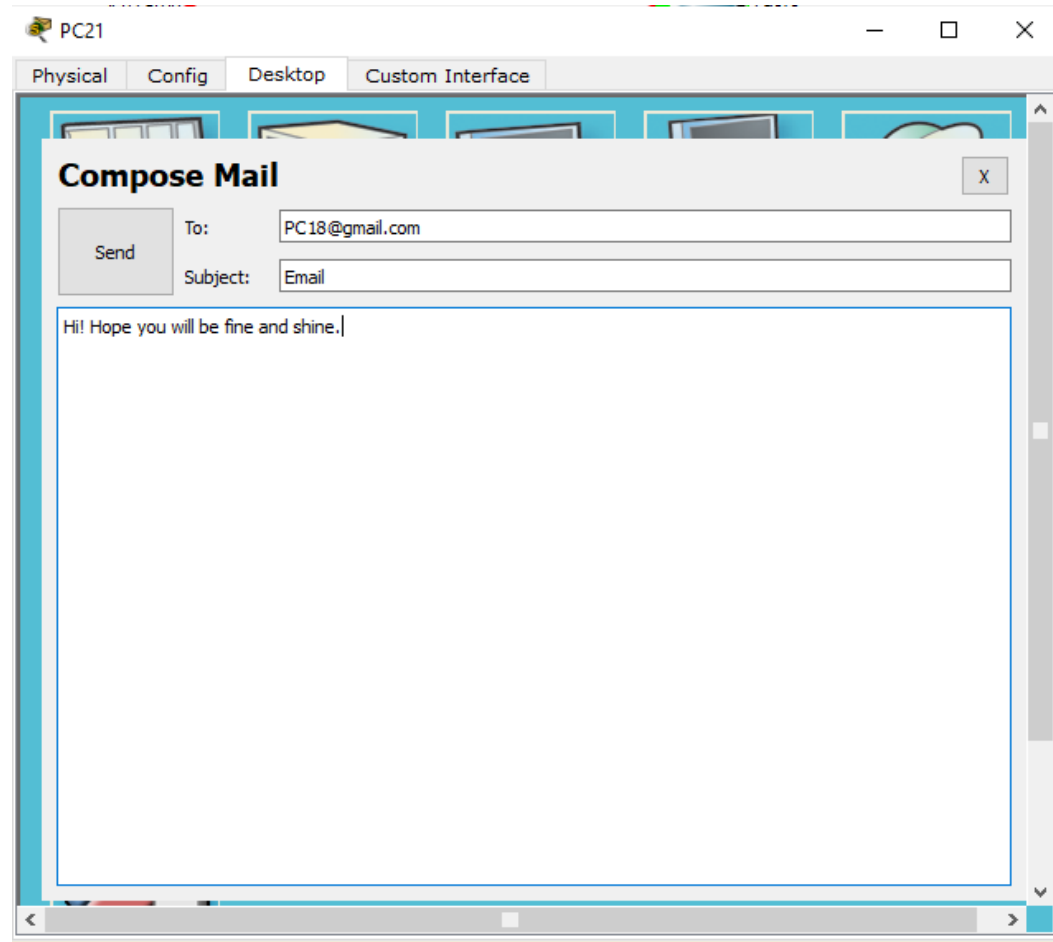
Logon Information

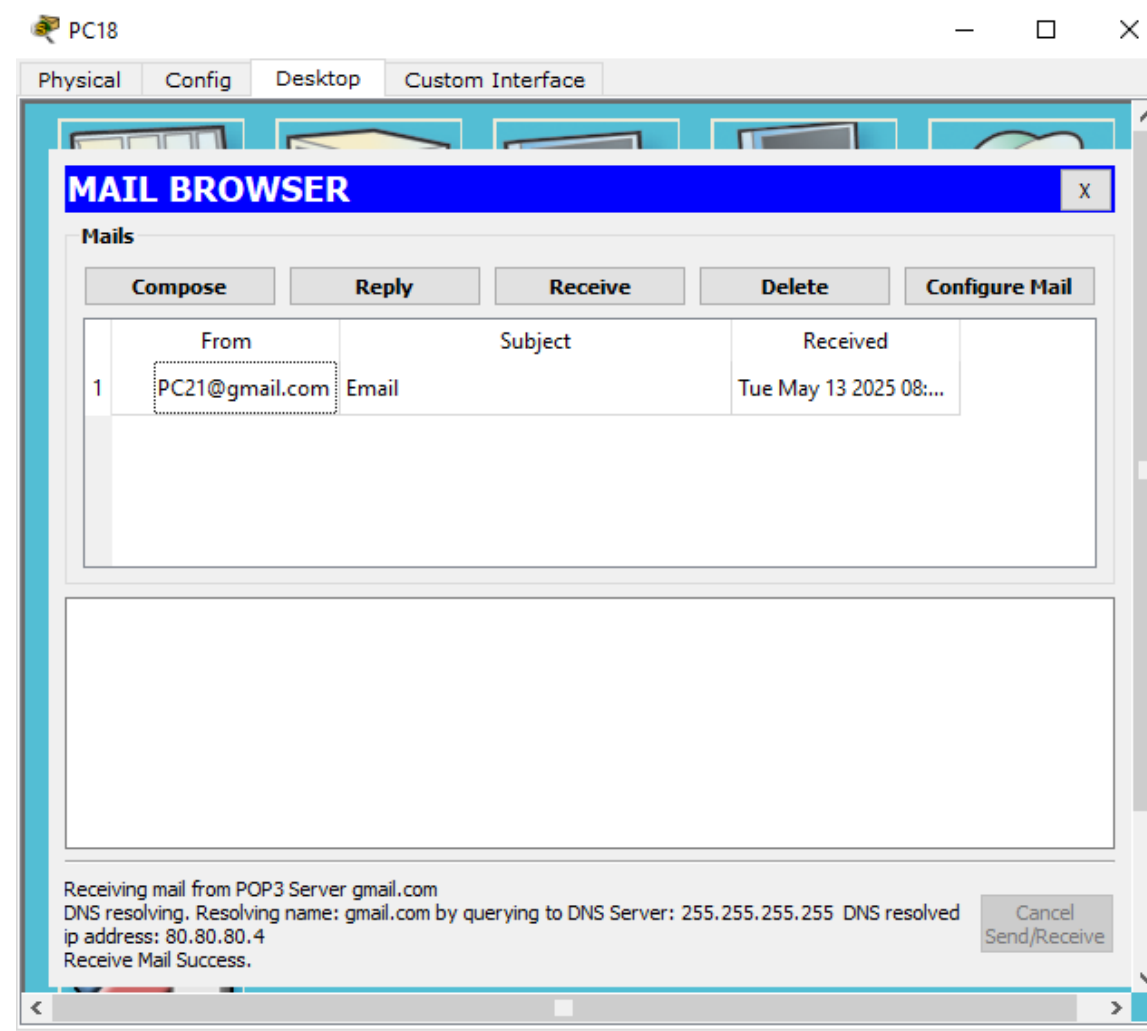
User Name: PC18

Password: ●●●

Save Clear Reset

PC21 to PC 18





PC18 to PC 21

